The Geometry of Parsimony: Spacetime as a Statistical Manifold equipped with a Fisher Information Metric

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The "Emergent Gravity" program suggests that spacetime geometry is not fundamental but arises from the entanglement structure of underlying quantum degrees of freedom. In this paper, we formalize this emergence using Information Geometry. We propose that the physical metric tensor $g_{\mu\nu}$ is isomorphic to the Fisher Information Metric defined on a statistical manifold of probability distributions. We demonstrate that the minimization of the Kullback-Leibler divergence (a measure of model complexity or "Parsimony") leads directly to the minimization of the Ricci scalar curvature. By identifying the system's "Context" with a holographic boundary, we recover the Einstein Field Equations from purely information-theoretic constraints, suggesting that Gravity is the geometric manifestation of efficient information processing.

I. INTRODUCTION

Since the discovery of Black Hole Thermodynamics [1], the link between information and geometry has been a central puzzle in physics. Jacobson [2] famously derived the Einstein equations from the Clausius relation $\delta Q = T\delta S$, suggesting that gravity is an equation of state.

More recently, the AdS/CFT correspondence implies that bulk geometry ("Parsimony") is dual to a boundary field theory ("Context") [3].

In this paper, we unify these perspectives under the PCIR framework. We argue that the "Parsimony" principle—the drive to minimize model complexity—is geometrically realized as the flattening of the statistical manifold. We show that the Einstein-Hilbert action is essentially a measure of information density.

II. THE STATISTICAL MANIFOLD

We consider a smooth manifold $\mathcal{M}$ where each point $\theta \in \mathcal{M}$ represents a probability distribution $P(x|\theta)$. In Information Geometry, such a manifold is naturally equipped with a unique Riemannian metric, the **Fisher Information Metric** (FIM), denoted here as $\mathcal{G}_{\mu\nu}$:

$$\mathcal{G}_{\mu\nu}(\theta) = \int dx\, P(x|\theta) \frac{\partial \ln P}{\partial \theta^\mu} \frac{\partial \ln P}{\partial \theta^\nu} \tag{1}$$

This metric measures the distinguishability of neighboring states. In the PCIR framework, we identify physical spacetime coordinates x^μ with the statistical parameters θ^μ.

III. PARSIMONY AS CURVATURE

A central tenet of Bayesian inference is the "Occam's Razor" or Parsimony principle: models should be as simple as possible. In geometric terms, "complexity" is often associated with the curvature of the manifold.

We verify this intuition by examining the volume element. The volume of distinguishable states in a region is given by the Jeffrey's prior measure $\sqrt{\det \mathcal{G}}$. [Image of curved spacetime manifold]

We postulate that the "Parsimony" cost function $\mathcal{C}_P$ is the integrated scalar curvature R of the Fisher metric over the manifold:

$$\mathcal{C}_P[\mathcal{G}] = \int_{\mathcal{M}} d^4x \sqrt{-\det \mathcal{G}}\, (R - 2\Lambda) \tag{2}$$

This is identically the Einstein-Hilbert action S_{EH}. Thus, General Relativity can be reinterpreted as the requirement that the statistical manifold minimizes its mean curvature—i.e., it seeks the "flattest" (simplest) geometry consistent with boundary conditions.

IV. CONTEXT: THE BOUNDARY CONDITION

To minimize this action, we require a boundary condition. In the PCIR framework, this is the **"Context" (C)**, physically realized as a holographic screen or Markov Blanket $\partial\mathcal{M}$.

Following the variational principle, we vary the metric $\mathcal{G}_{\mu\nu}$ with respect to the action:

$$\delta \mathcal{C}_P = \int_{\mathcal{M}} d^4x \sqrt{-\mathcal{G}} \left(R_{\mu\nu} - \frac{1}{2} R \mathcal{G}_{\mu\nu} + \Lambda \mathcal{G}_{\mu\nu} \right) \delta \mathcal{G}^{\mu\nu} + \oint_{\partial\mathcal{M}} \dots \tag{3}$$

The boundary term (Context) encodes the entanglement entropy of the system with its environment. If we associate the variation of information on the boundary with the stress-energy tensor $T_{\mu\nu}$ in the bulk (via the Ryu-Takayanagi formula [4]), we enforce the constraint:

$$R_{\mu\nu} - \frac{1}{2} R \mathcal{G}_{\mu\nu} + \Lambda \mathcal{G}_{\mu\nu} = \kappa T_{\mu\nu} \tag{4}$$

This recovers the Einstein Field Equations.

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V. INTERPRETATION

This derivation provides a rigorous definition for the "P" and "C" components of the framework:

- **Parsimony (P):** The Einstein-Hilbert Action. The universe minimizes the scalar curvature of its statistical manifold. Gravity is the "restoring force" that tries to keep the geometry simple (flat) against the disturbing influence of matter.

- **Context (C):** The Boundary Terms. The geometry is determined by the information flux (entanglement) across the horizon or Markov Blanket.

Matter ($T_{\mu\nu}$) acts as a source of "complexity" or "surprise," curving the manifold and forcing the geometry to deviate from maximum parsimony.

VI. CONCLUSION

We have demonstrated an isomorphism between the Fisher Information Metric and the spacetime metric tensor. This supports the hypothesis that Spacetime is an emergent phenomenon arising from the statistical properties of a large number of degrees of freedom.

In conjunction with Paper 2 (Renormalization/Dynamics) and Paper 3 (Quantum Inference), this completes the static geometric description of the PCIR framework.

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